



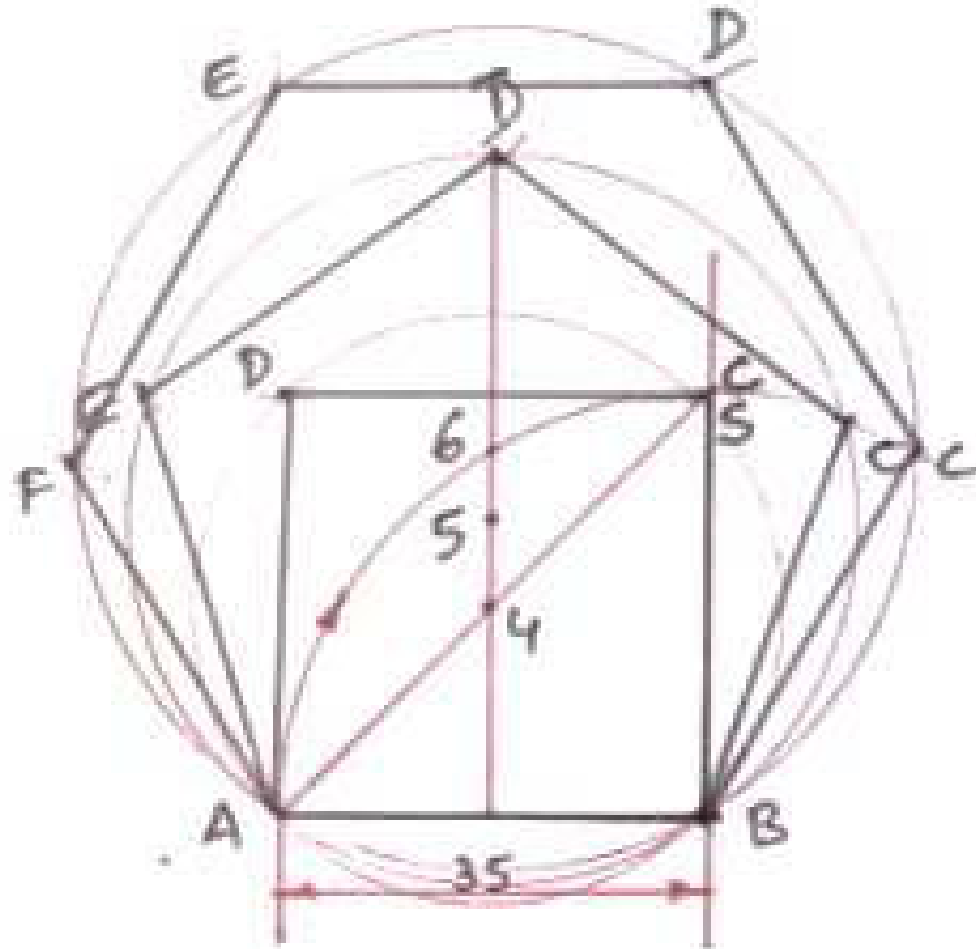
Complete Paper Solution

EGD – Summer 2024

Engineering Graphics and Design

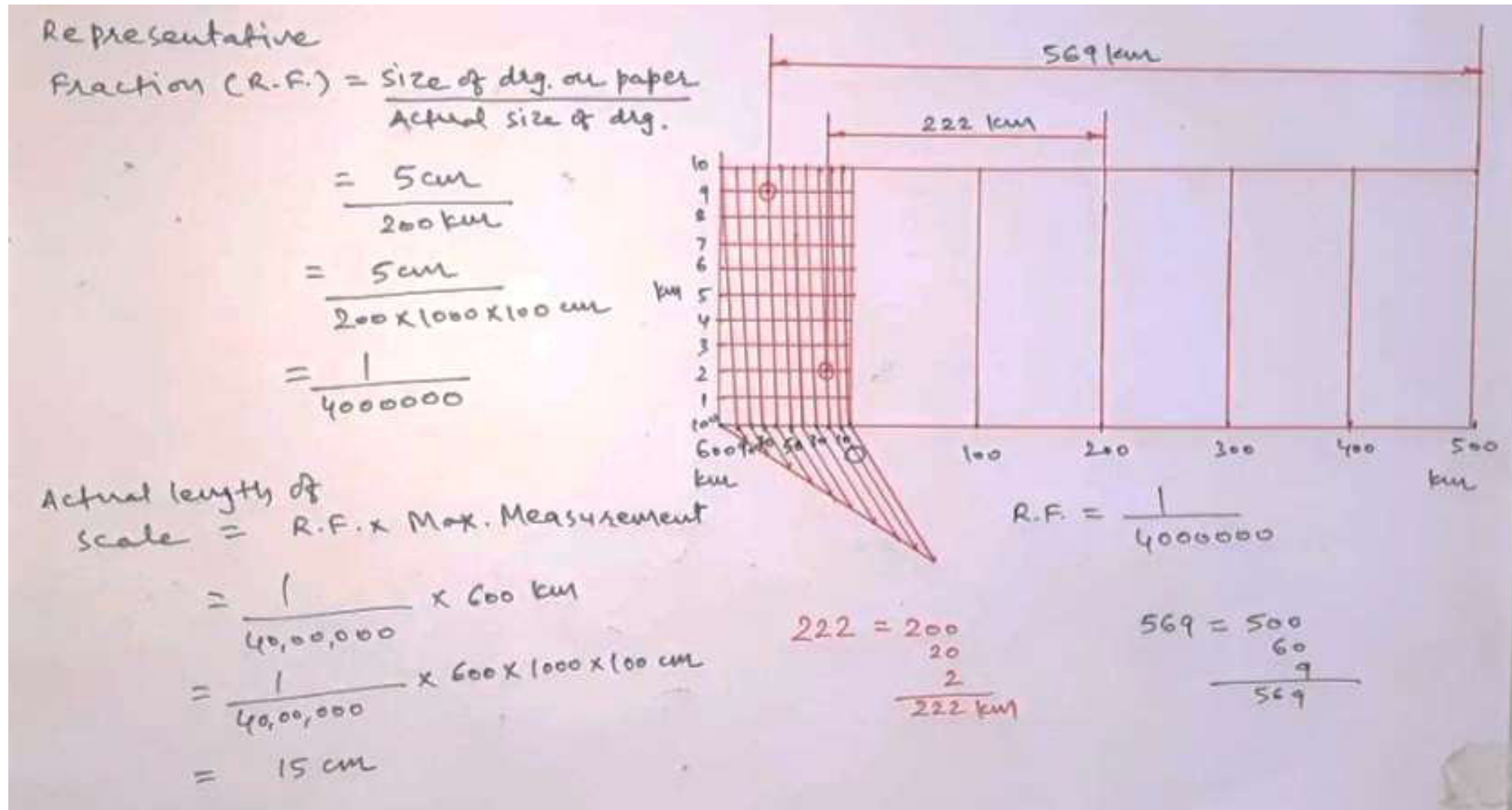
(GTU)

Q. 1 (a) Draw regular square, pentagon and hexagon by UNIVERSAL METHOD of 35 mm side. **(3 Marks)**



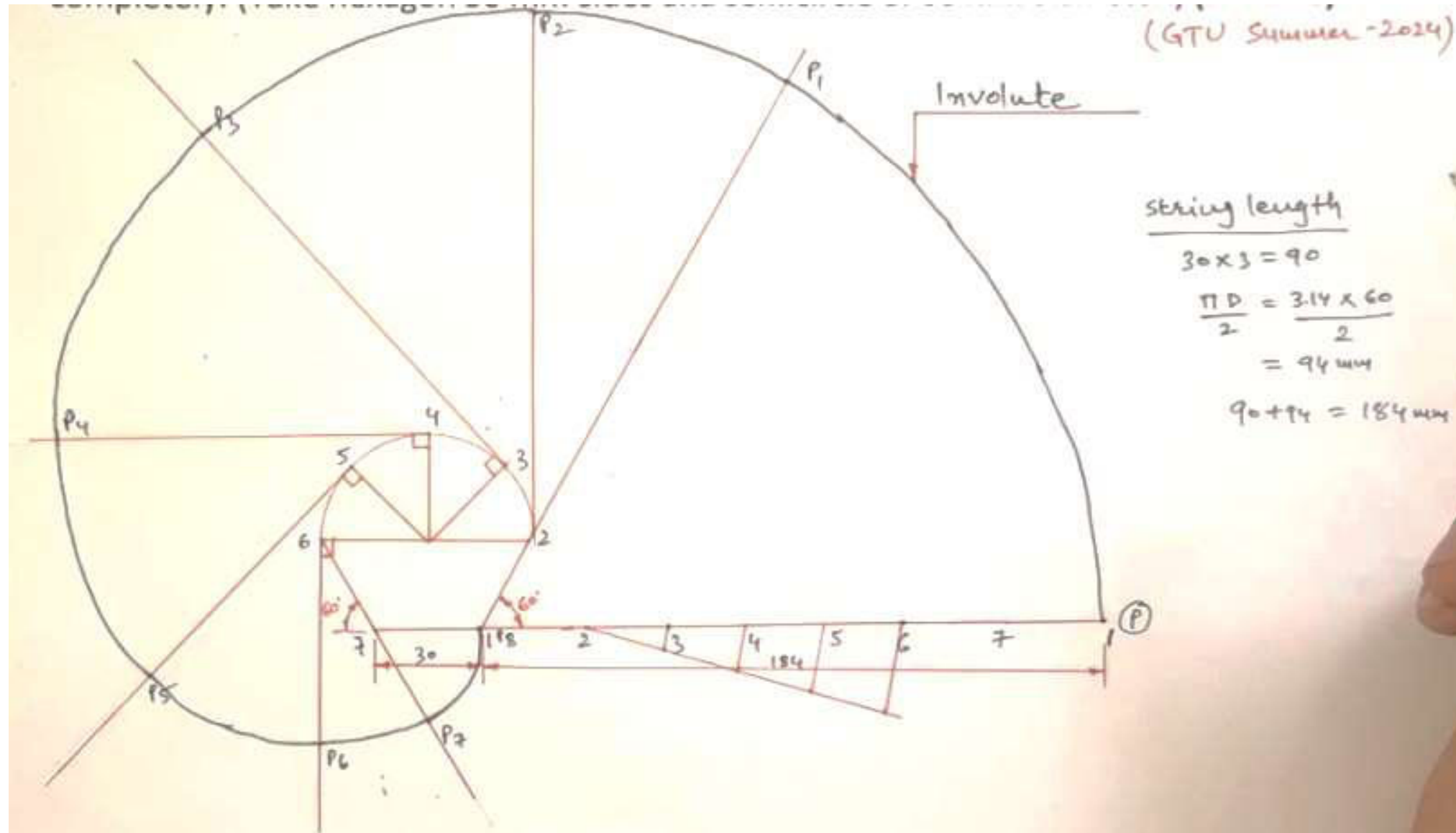


Q. 1 (b) The distance between Ahmedabad and Surat is 200 km. In a railway map it is represented by a line 5 cm long. Find it's R.F. Draw a diagonal scale to show single km. And maximum 600 km. Indicate on it following distances. (1) 222 km (2) 569 km. **(4 Marks)**

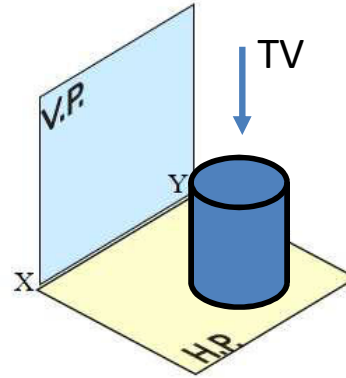




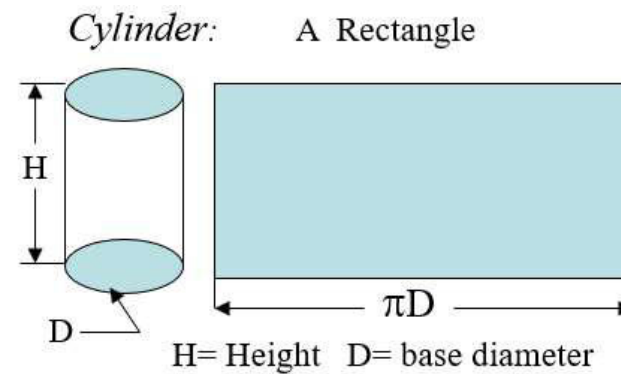
Q. 1 (c) A pole is of a shape of half hexagon and semicircle. A string is to be wound having length equal to the pole perimeter. Draw path of free end P of string when wound completely. (Take hexagon 30 mm sides and semicircle of 60 mm diameter.) **(7 Marks)**



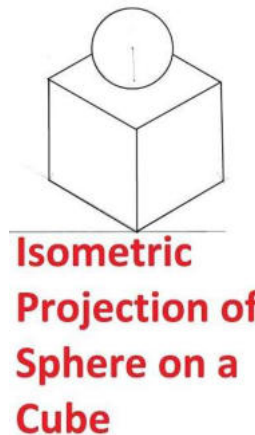
Q. 2 (a) (1) The top view of a right cylinder resting on HP on its base is **Circle.**
(3 Marks)



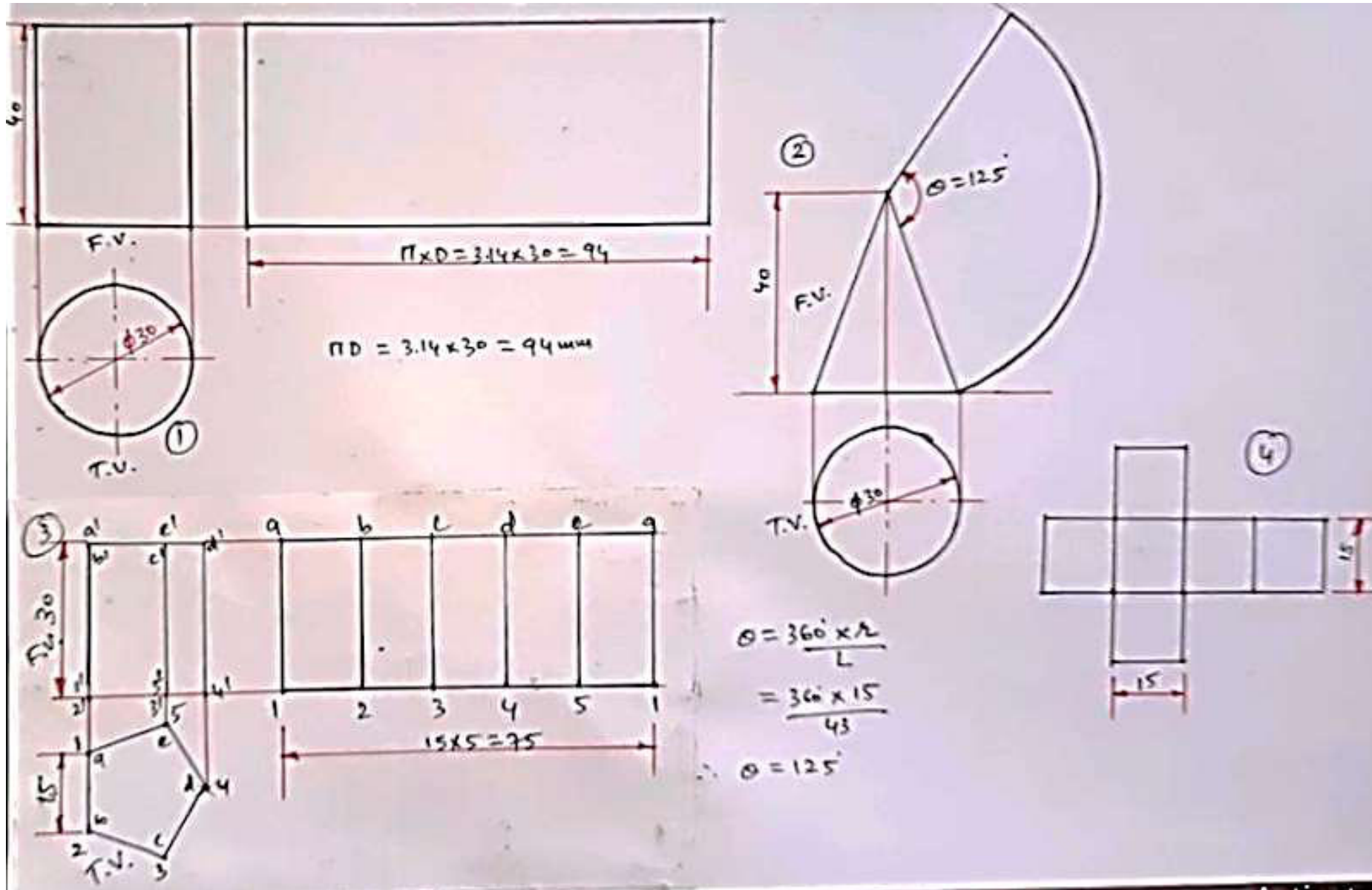
(2) The development of cylinder is a **Rectangle**



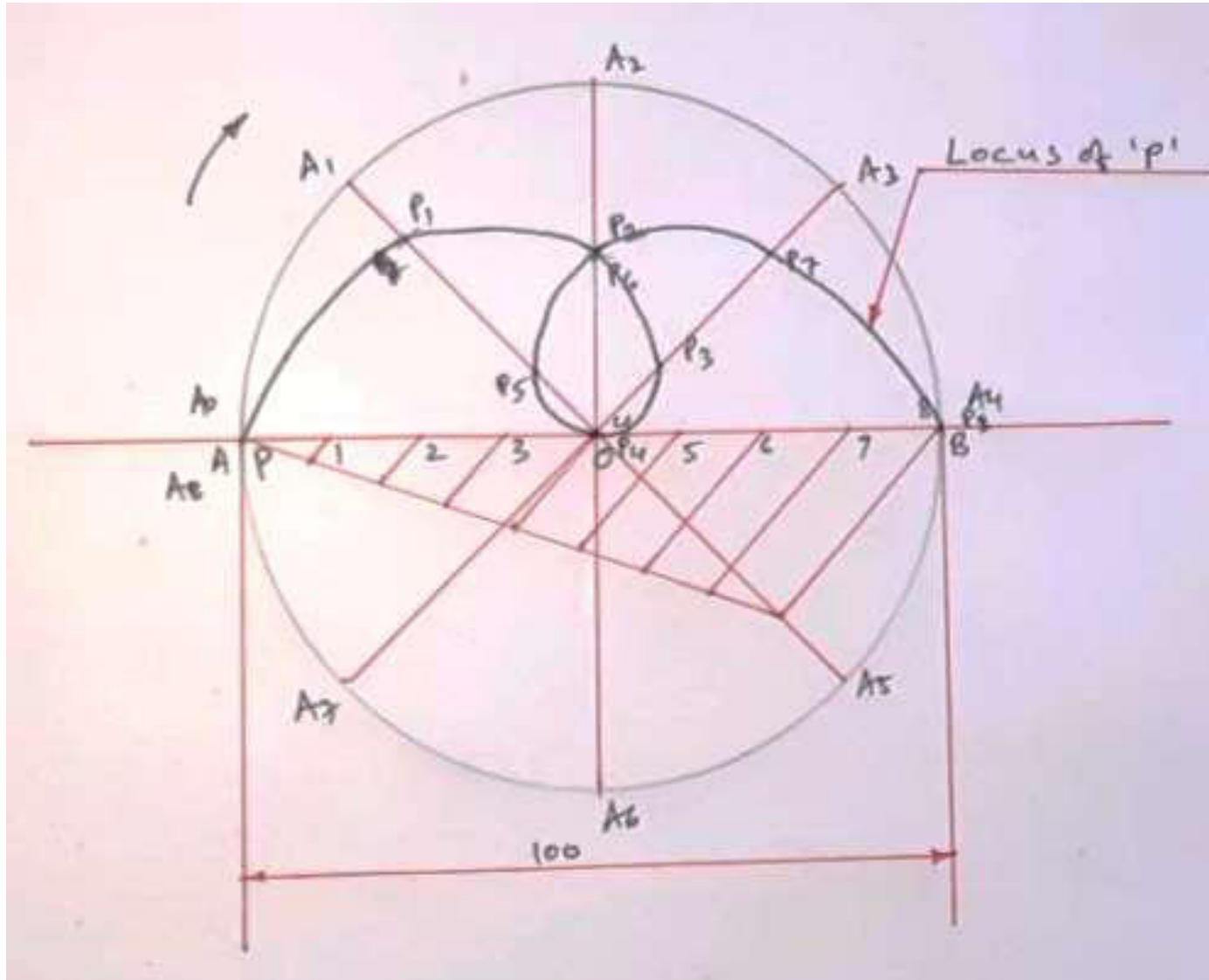
(3) The isometric projection of a sphere is a **Circle.**



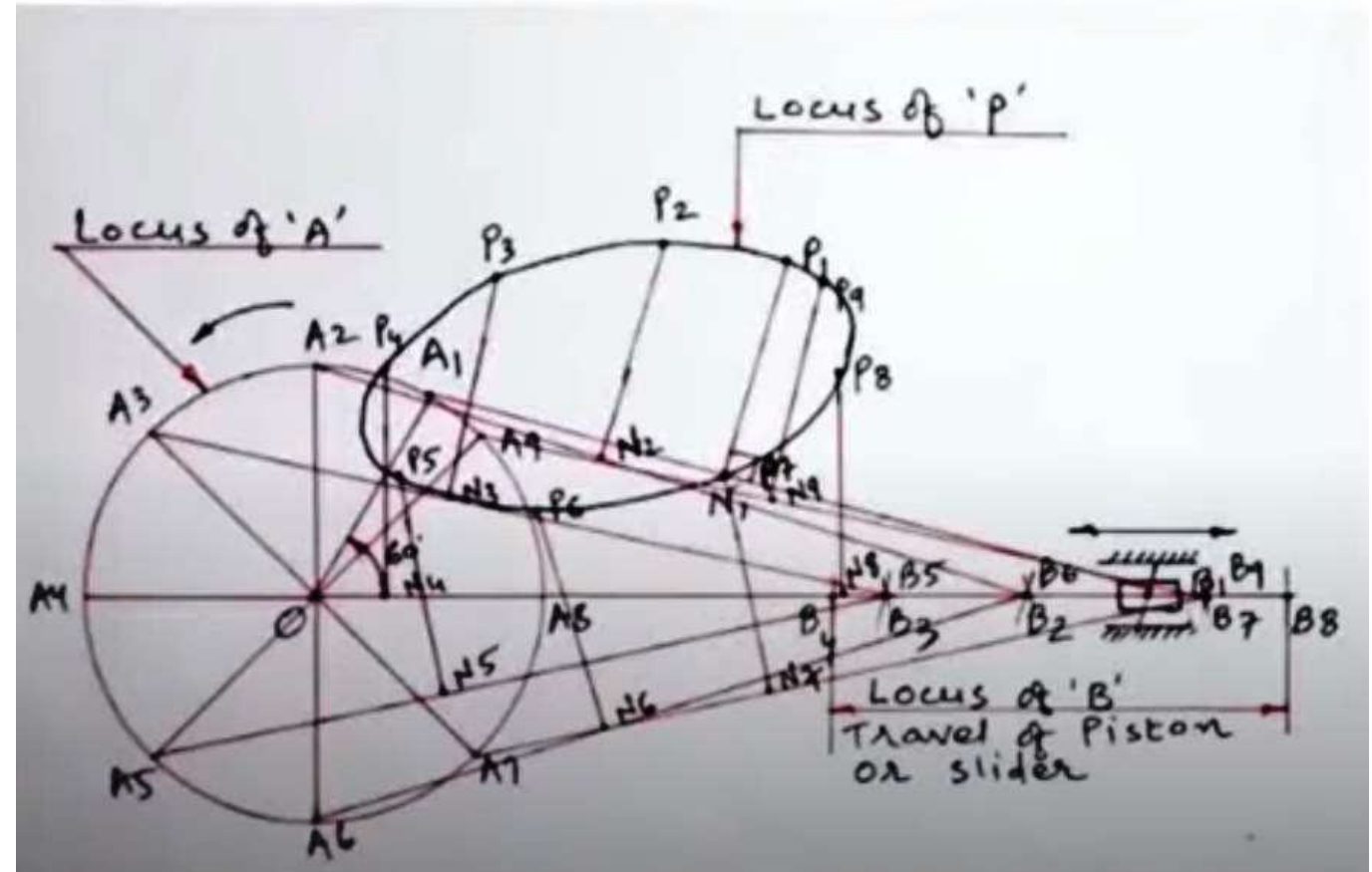
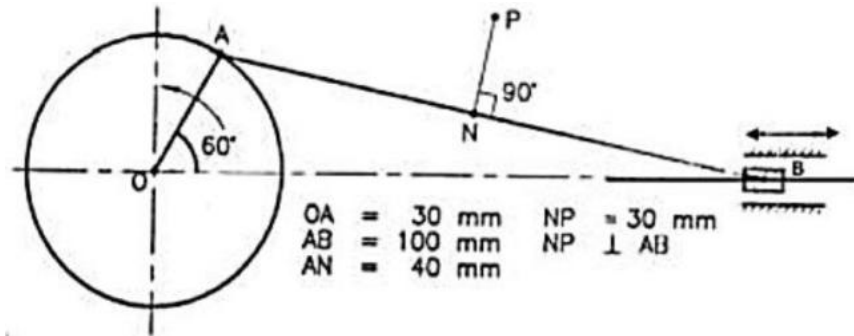
Q. 2 (b) Draw Development of lateral surfaces of following solids.
 (1) Cylinder (2) Cone (3) Pentagonal Prism (4) Cube **(4 Marks)**



Q. 2 (c) Rod AB, 100 mm long, revolves in clockwise direction for one revolution. Meanwhile point P, initially on A starts moving towards B and reaches B. Draw locus of point P. **(7 Marks)**



Q. 2 (c) OR Draw loci of point 'P' given in figure (7 Marks)



Video Link: <https://youtu.be/L1YgqM863rw>





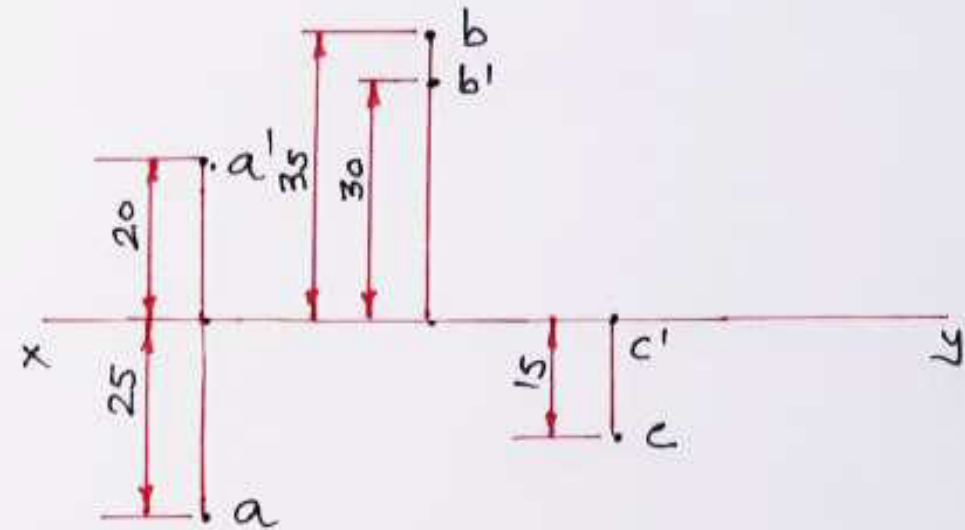
Q. 3 (a) Draw following projection of points:

(1) Point 'A' is 20 mm above HP and 25 mm In front of VP.

(2) Point 'B' is 35 mm behind VP and 30 mm above HP.

(3) Point 'C' is in the HP and 15 mm In front of VP. **(3 Marks)**

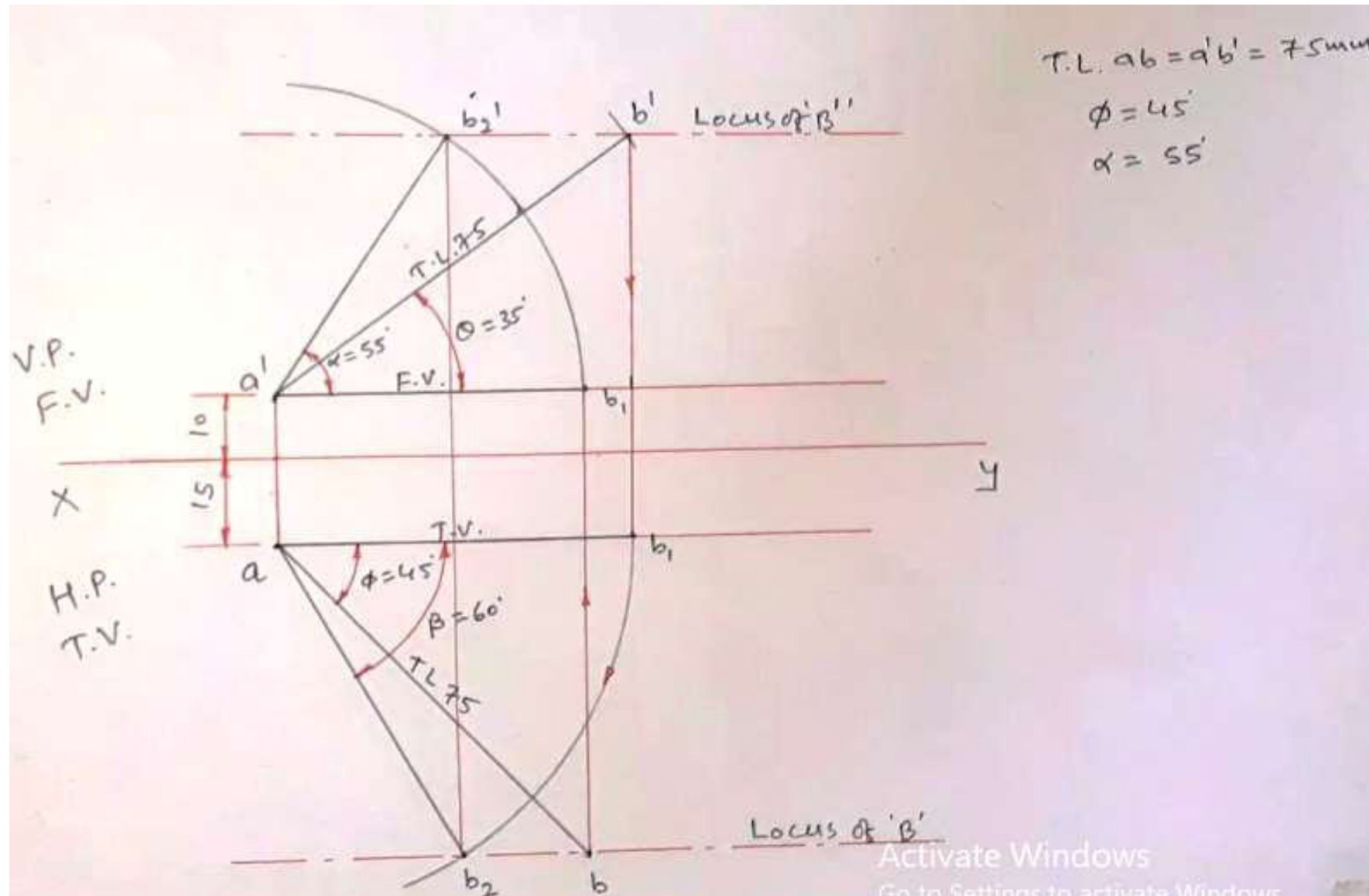
Point A above H.P.	
Point A below H.P.	
Point A on/in the H.P.	
Point A behind V.P.	
Point A in front of V.P.	
Point A on/in the V.P.	







Q. 3 (c) Line AB 75mm long makes 45° inclinations with VP while it's FV makes 55° . End A is 10 mm above HP and 15 mm in front of VP. If line is in 1st quadrant draw its projections and find it's inclination with HP. **(7 Marks)**

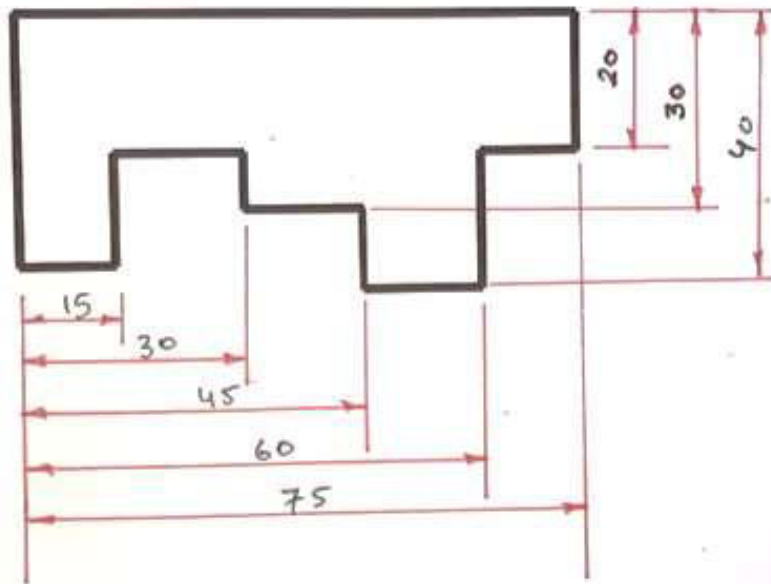


T.L. $ab = a'b' = 75 \text{ mm}$
 $\phi = 45^\circ$
 $\alpha = 55^\circ$

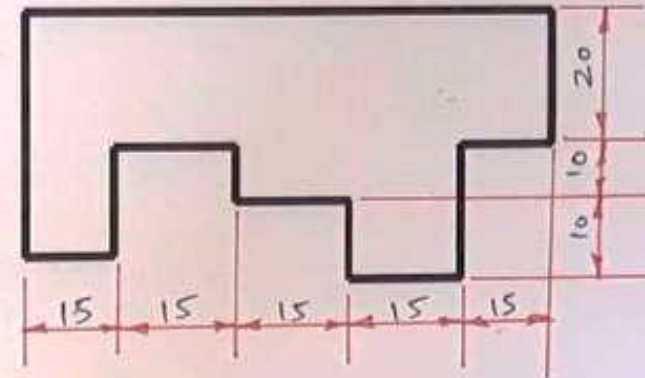
**Q. 3 (a) OR Draw Parallel, Chain and Combined Dimensioning for following figure.
(3 Marks)**



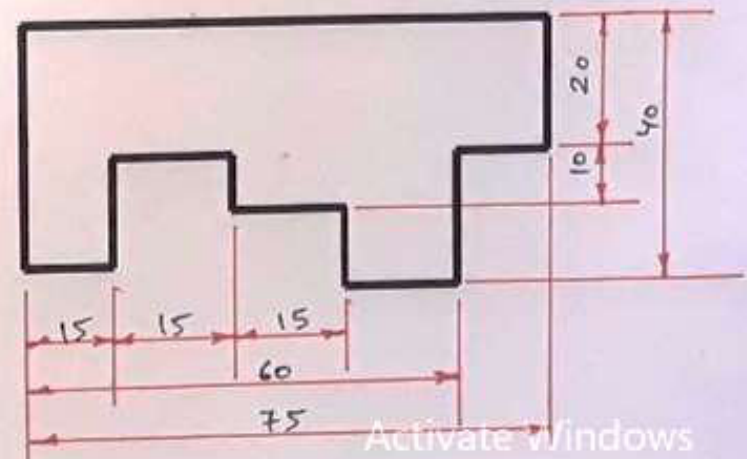
Parallel Dimensioning



Chain Dimensioning



Combined Dimensioning



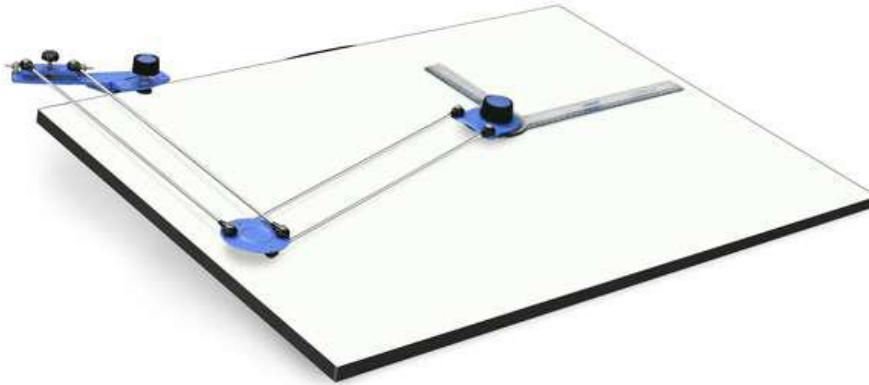
Activate Windows

Go to Settings to activate Windows



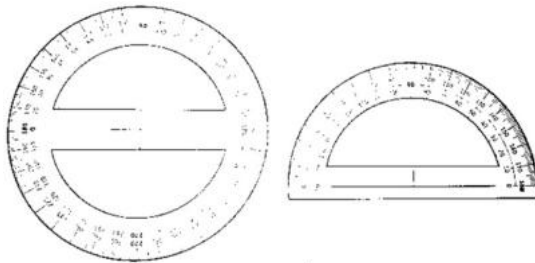
Q. 3 (b) OR Write at least one use of following drawing instruments: **(4 Marks)**

(1) Mini drafter :



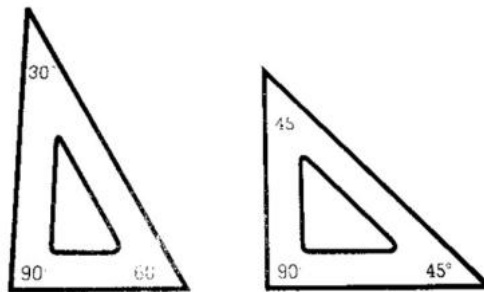
It is used to draw faster. It serves the purpose of T-square, set square, protractor and scale.

(2) Protector :



it is used for the measurement of angles.

(3) Set squares :



Set squares are used with T – square to draw lines at various angles.

(4) Compass :



It is used to draw circles and arcs.

(5) Divider :



It is like a compass except it has metal points at both legs. It is used to transfer distances or dimensions and sometimes for dividing space into equal divisions.

(6) Roll-N-Draw:



It is used to draw parallel line quickly and easily.



(7) T-square :



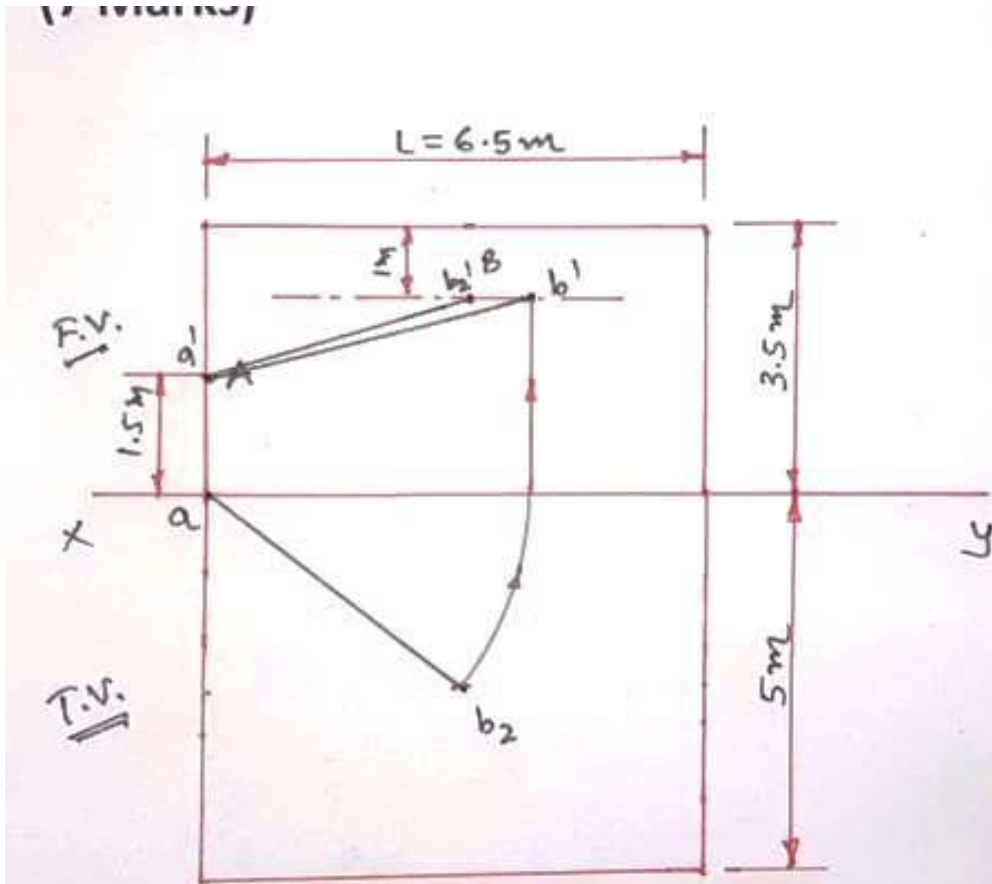
The T – square slides up and down along the straight working edge of the board to draw parallel horizontal straight lines.

(8) French curve :

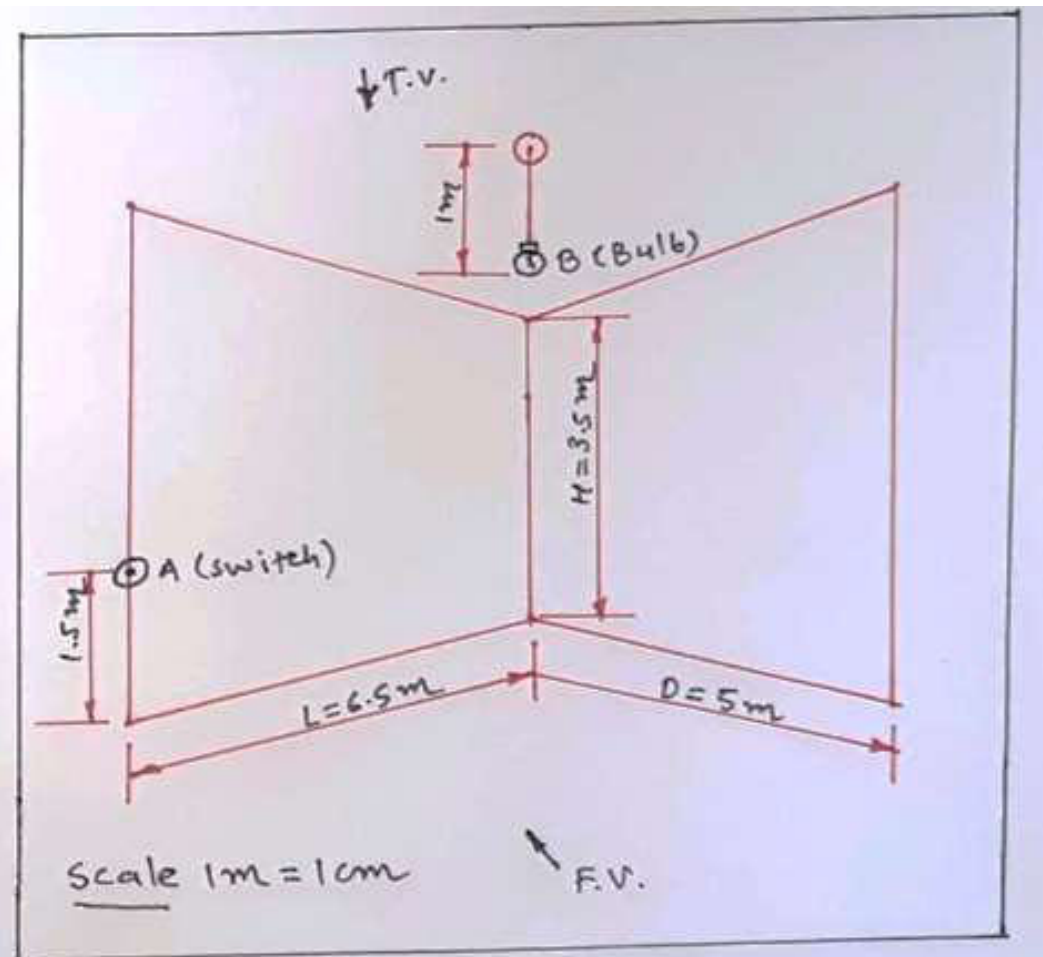


It is used to draw curves other than arc of circles accurately and uniformly. i.e. it is used to draw irregular arcs.

Q. 3 (c) OR A room is of size 6.5m L, 5m D, 3.5m high. An electric bulb hangs 1m below the center of ceiling. A switch is placed in one of the corners of the room, 1.5m above the flooring. Draw the projections and determine real distance between the bulb and switch. (7 Marks)

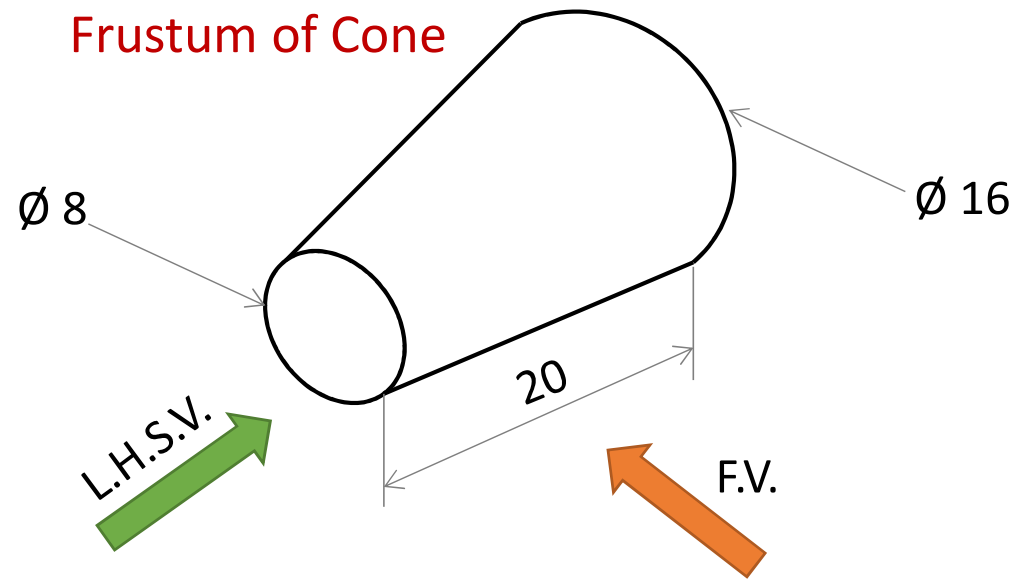


So, real distance between the bulb and switch is = 4.3m

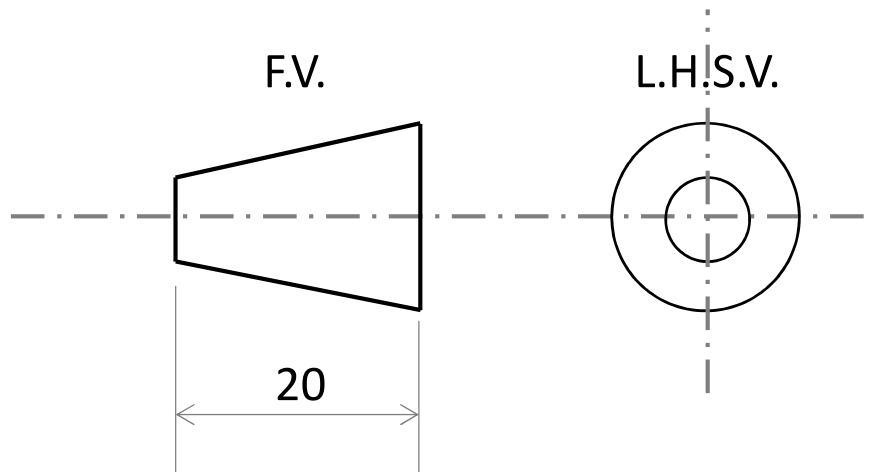


Activate Windows

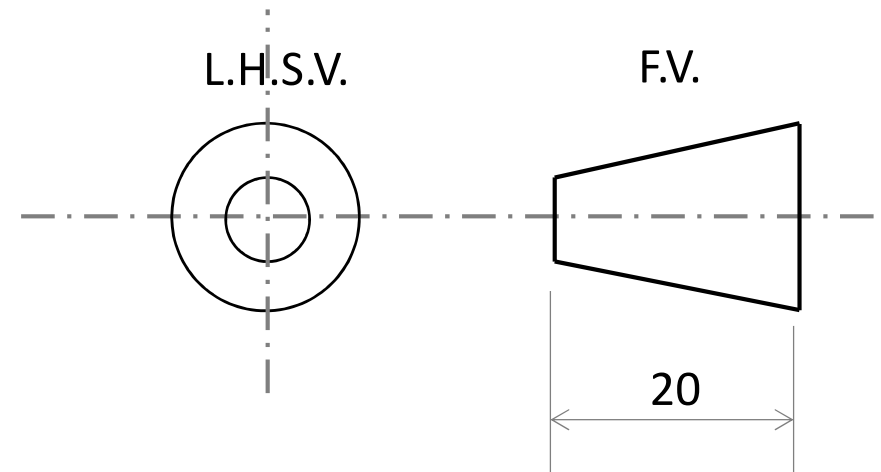
Q. 4 (a) Draw symbol of first angle and third angle projection method. **(3 Marks)**



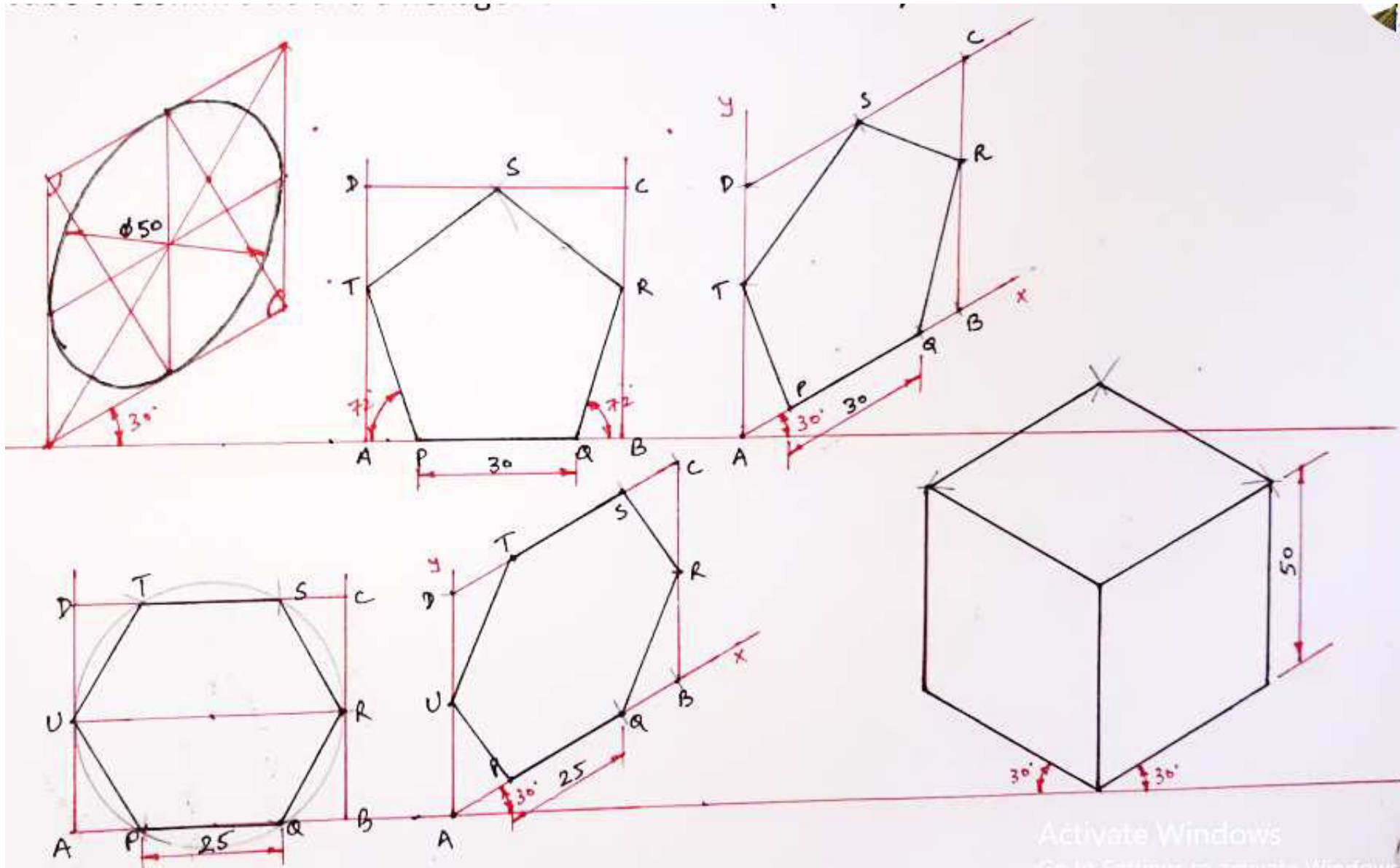
First Angle Projection Method



Third Angle Projection Method

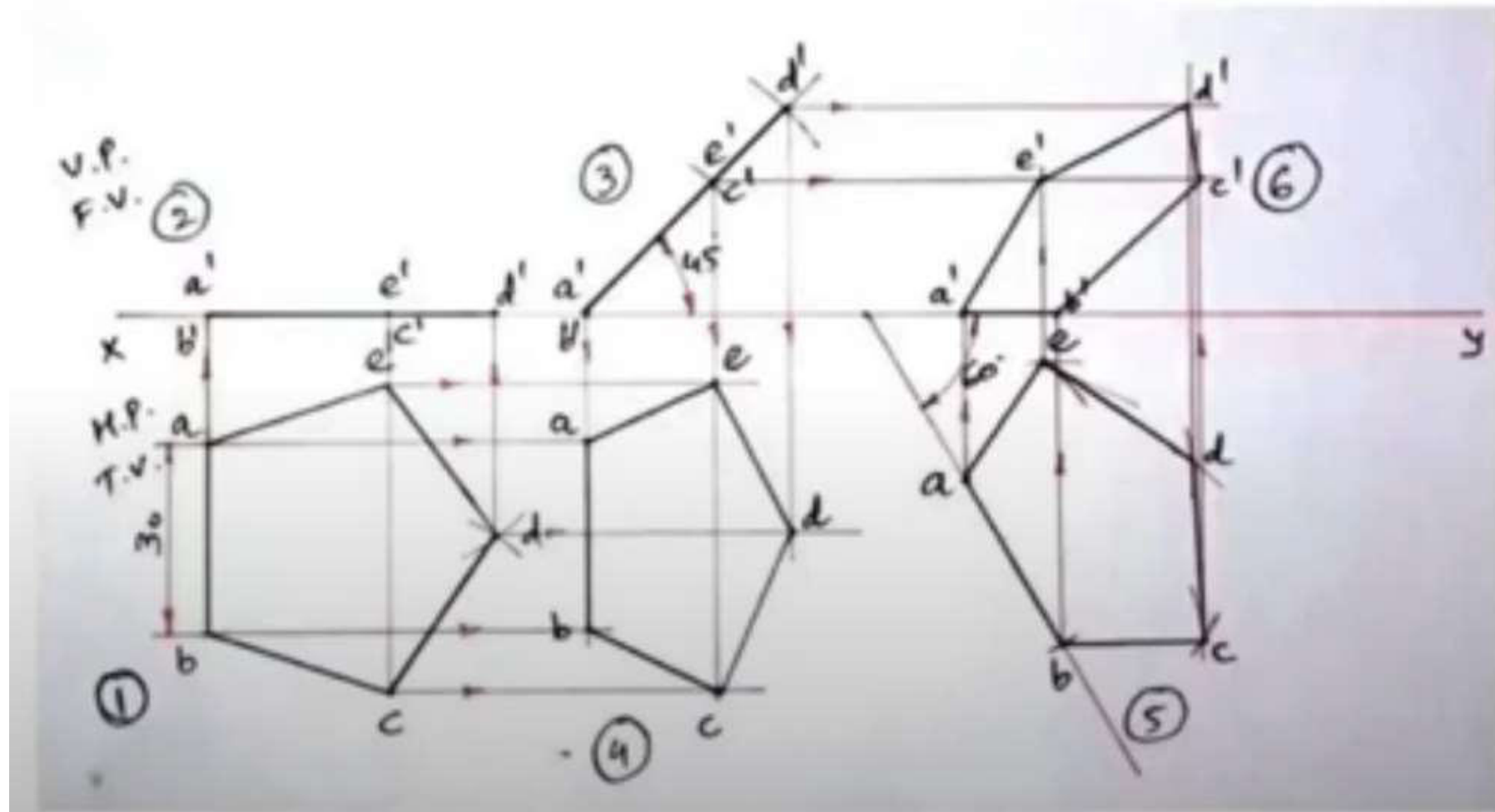


Q. 4 (b) Draw Isometric view of Circle of 50mm diameter, Pentagon of 30 mm side, a cube of 50mm side and a hexagon of 25 mm side. **(4 Marks)**





Q. 4 (c) A regular pentagon of 30 mm sides is resting on HP on one of its sides with its surface 45° inclined to HP. Draw its projections when the side in HP makes 60° angles with VP. **(7 Marks)**

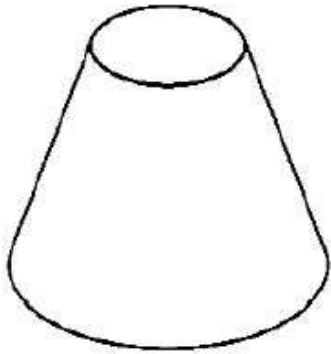


Video Link: <https://youtu.be/xnp5M9QfJLE>

Q. 4 (a) OR Identify following solids: (3 Marks)

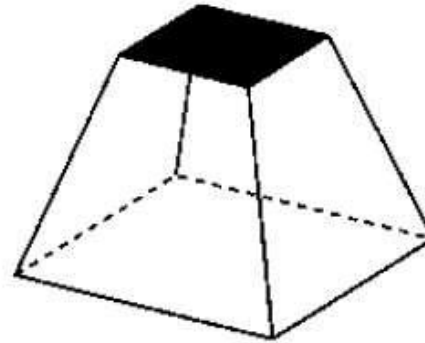


(1)



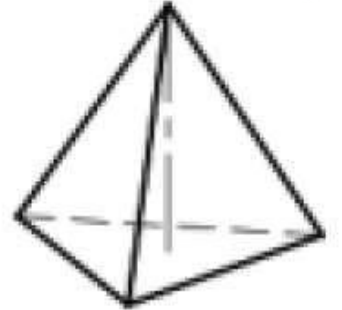
Frustum of Cone

(2)

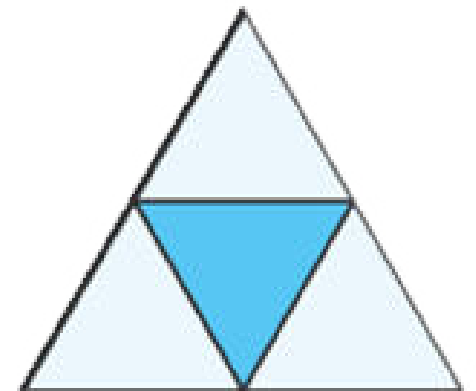
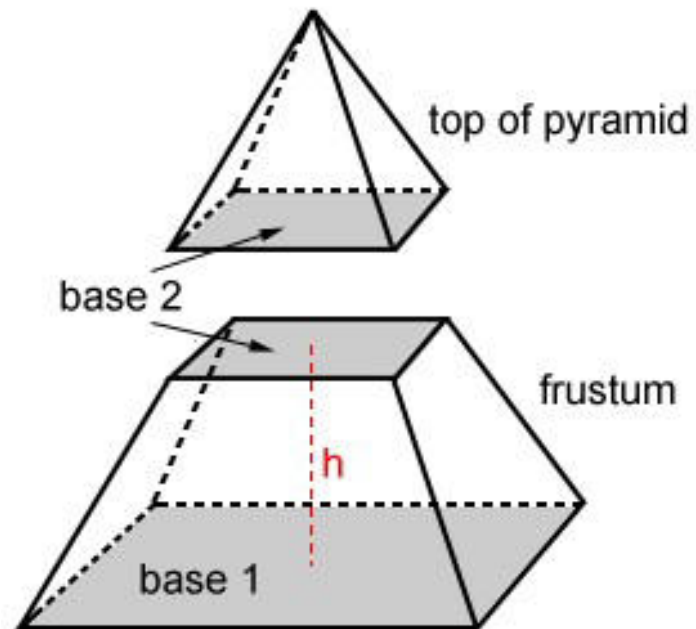
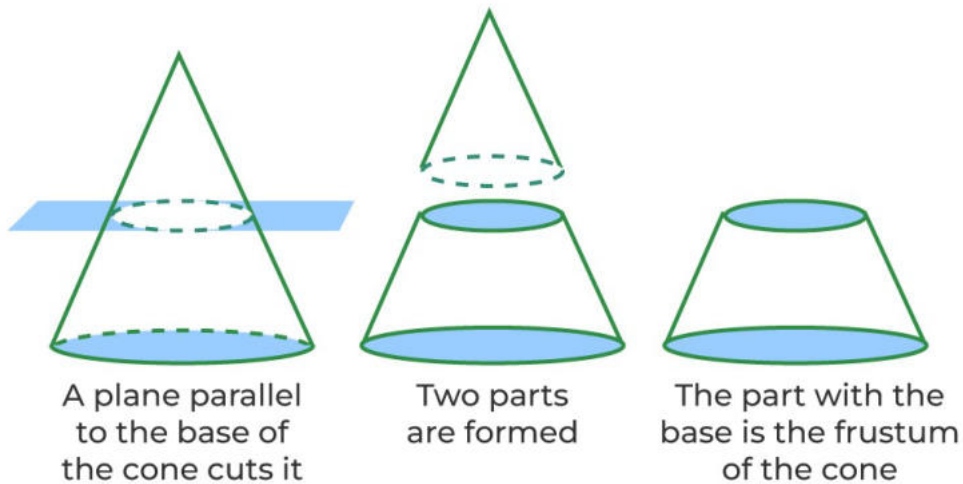


Frustum of Square Pyramid

(3)

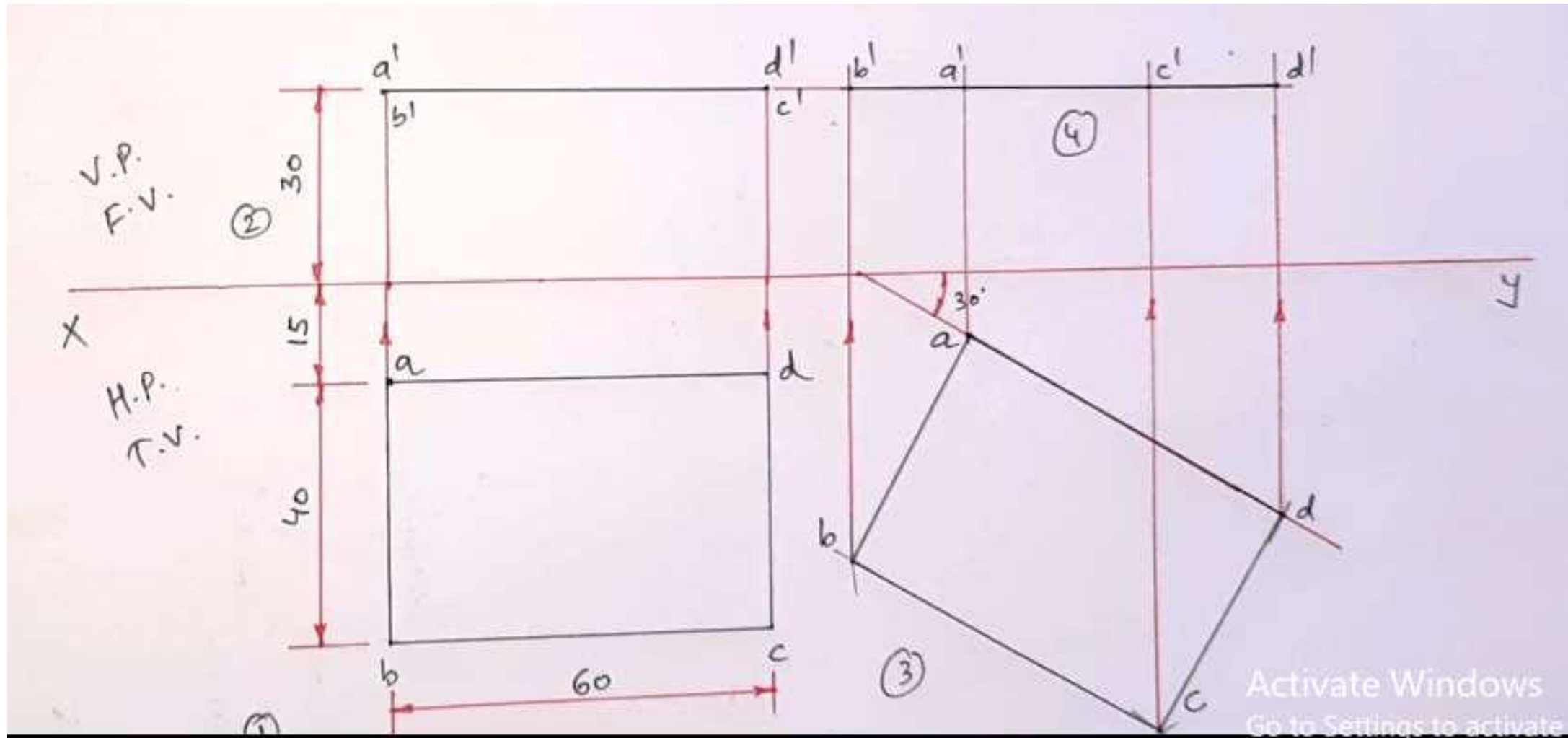


Tetrahedron
(Triangular Pyramid)

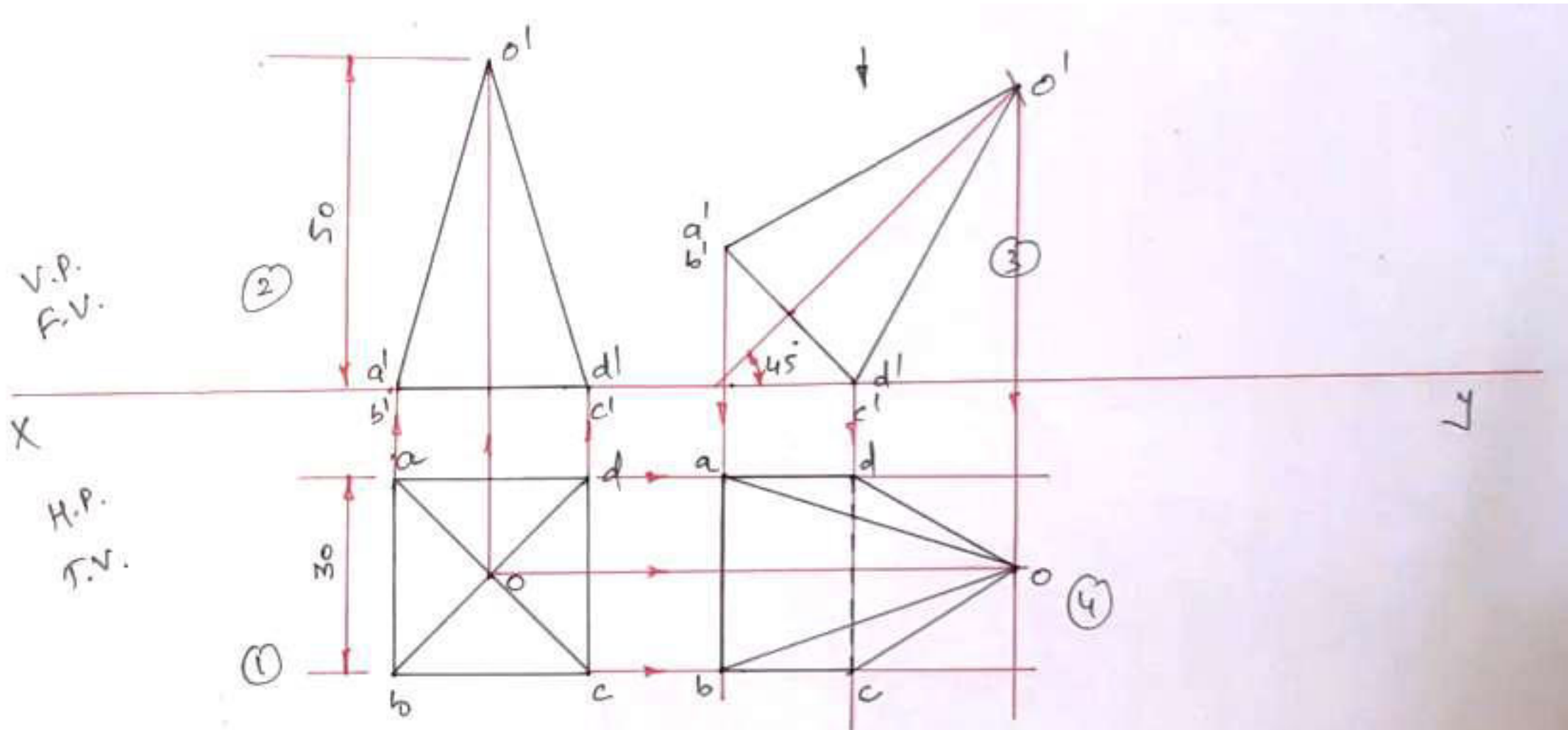




Q. 4 (b) OR A rectangle ABCD 60 mm x 40 mm, is parallel to HP with one of its sides inclined at 30° to VP and the end of the side near to VP is 15 mm in front of the VP and 30 mm above HP. Draw its projections. **(4 Marks)**



Q. 4 (c) OR A square pyramid, side of base 30 mm and height 50 mm, is resting on HP with a side of its base. Draw the projections if the axis is inclined to 45° to HP and parallel to VP (**7 Marks**)



Q. 5 (a) What is meant by ARRAY in AUTOCAD? What are types of ARRAYS? (4 Marks)



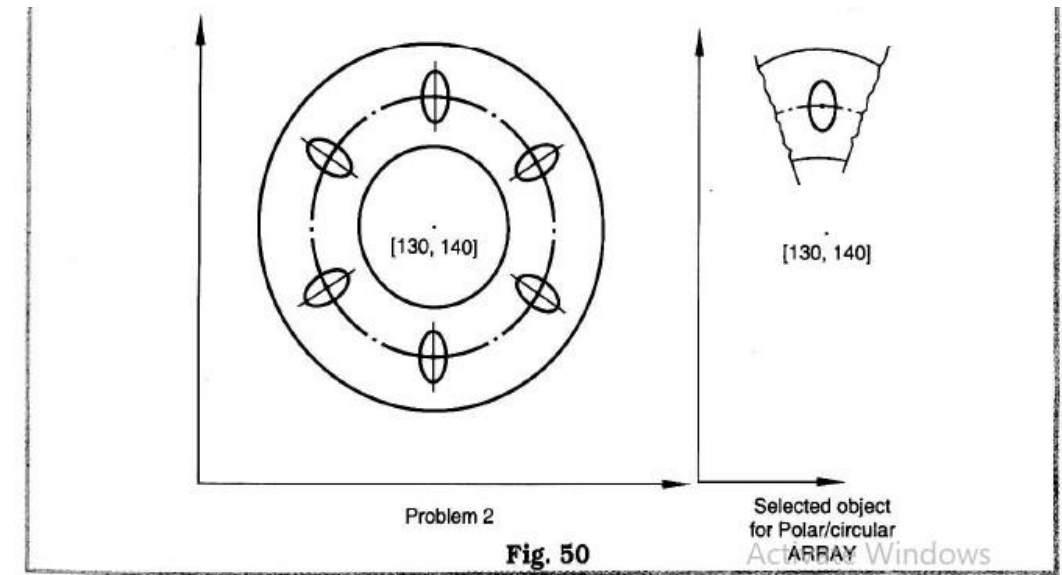
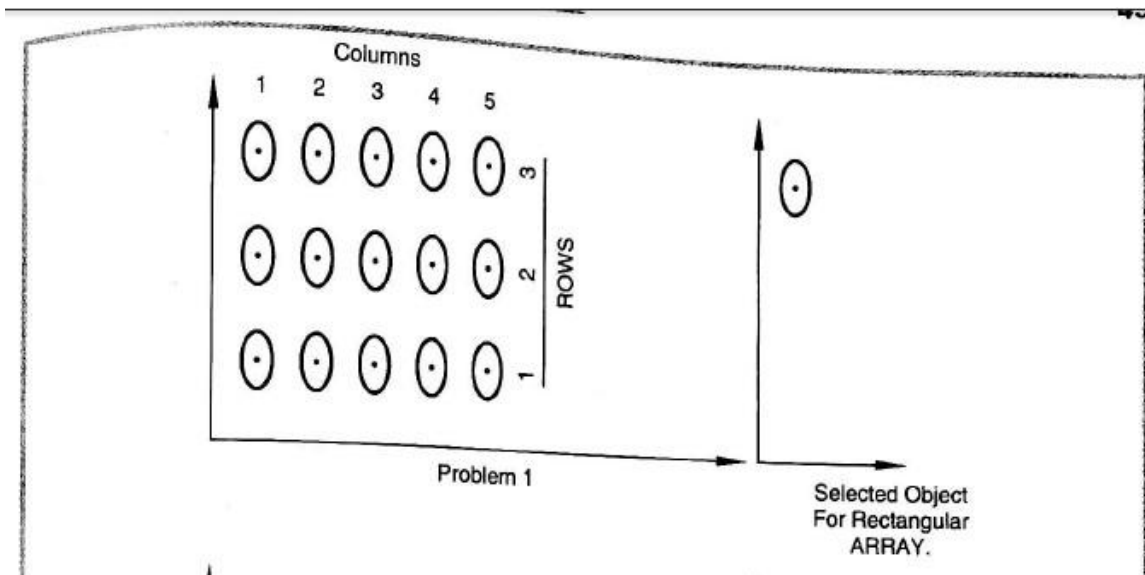
Array command is used to create multiple copies of selected object in a rectangular pattern or polar / circular pattern. So, there are two types of array,

1. Rectangular array

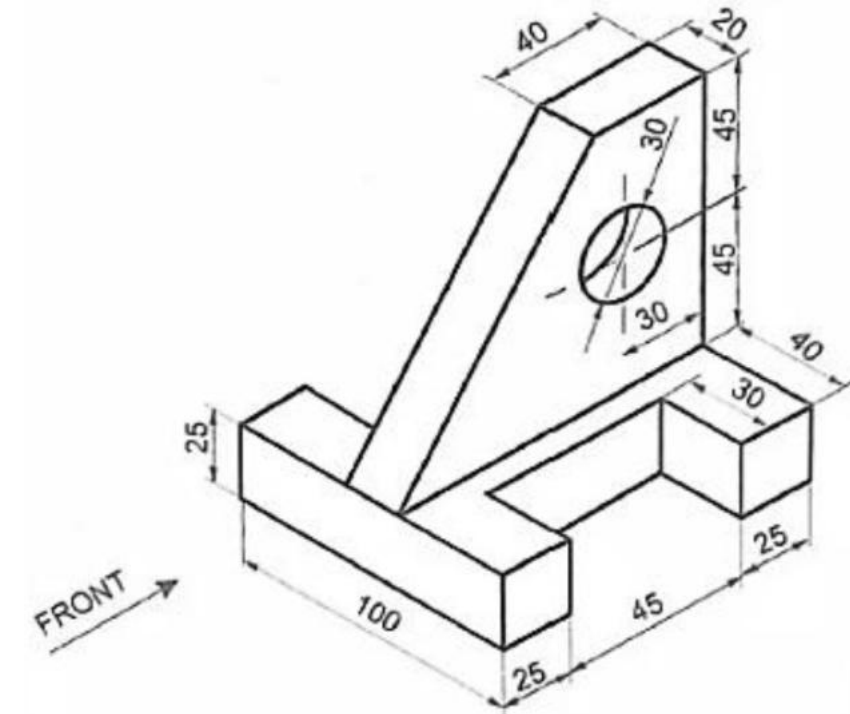
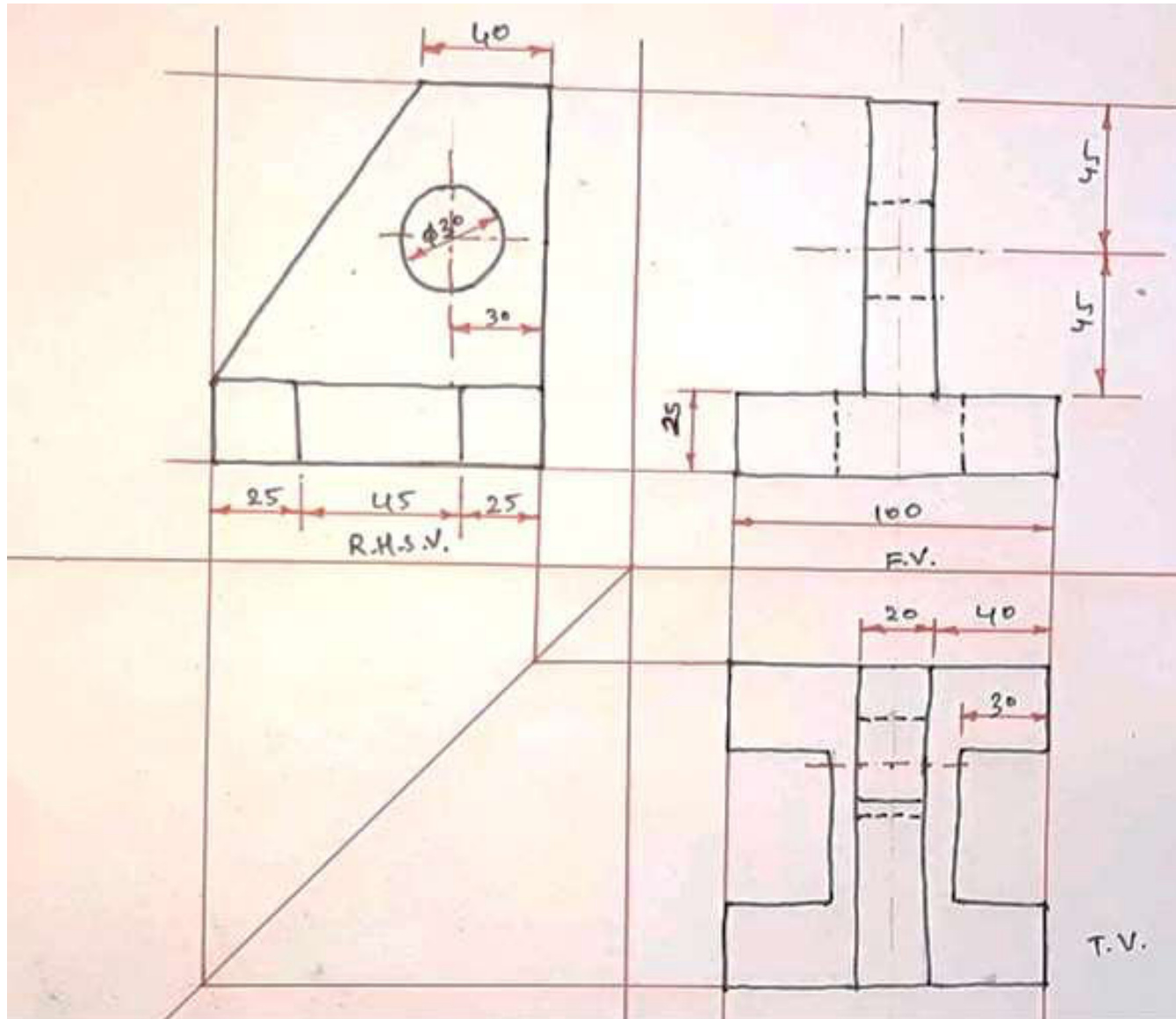
- It is used for seating arrangements in auditorium, tiles fitting in hall etc.

2. Polar / Circular array

- It is used for gear teeth, sprocket teeth etc.



Q. 5 (b) Draw orthographic projections of given figure (FV, TV, RHSV) as per first angle projection method. **(10 Marks)**



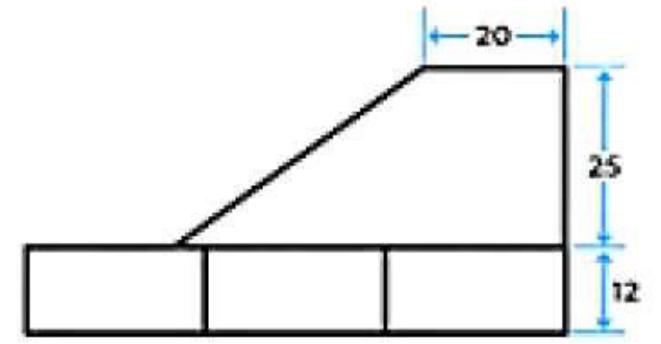
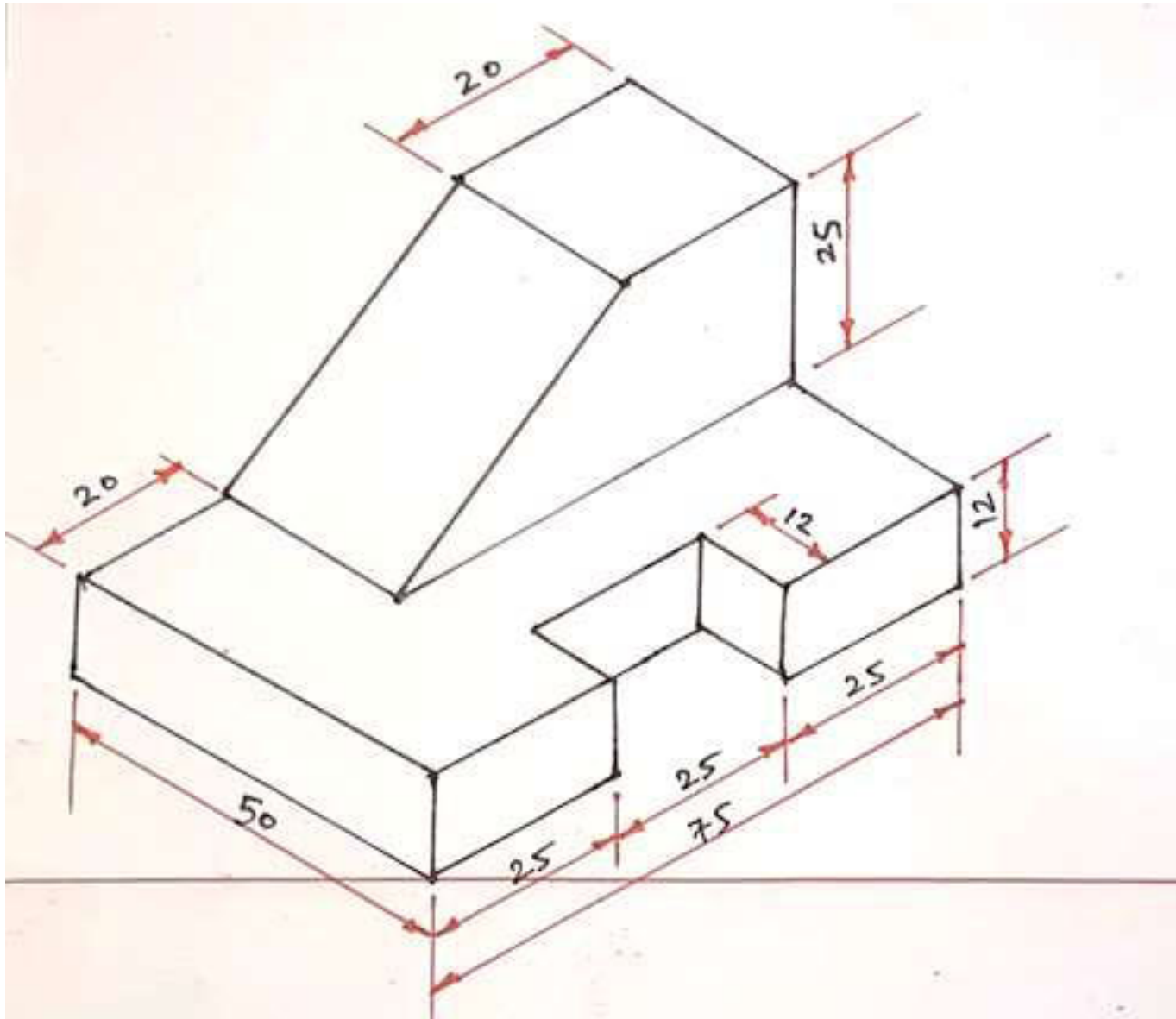


Q. 5 (a) OR Write down different software used in design. (4 Marks)

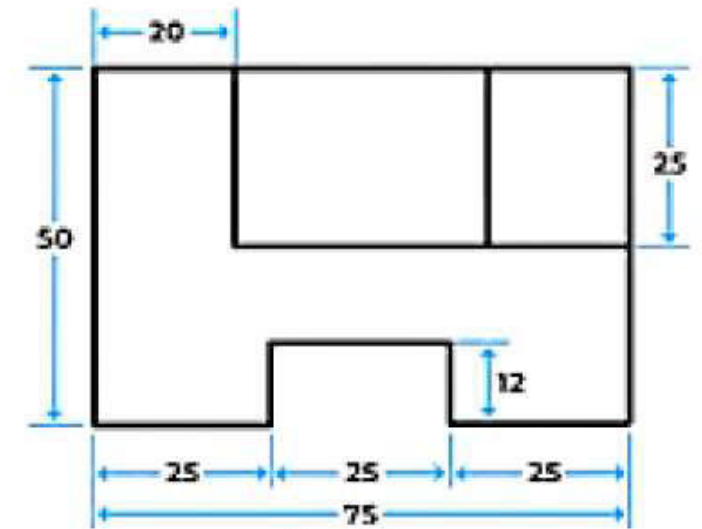
Different software used in design is as per the following.

1. AutoCAD
2. Solidworks
3. Creo
4. Catia
5. Fusion 360

Q. 5 (b) OR Draw isometric scale for length of 100mm. and draw isometric view for given figure. **(10 Marks)**



Front View



Top View